MES4-Carrier-EC · design review package

Input for review R3 · schematic Rev A, 28 Oct 2026 (branch `mes4-gpio-standard`) · Monkey Robotics · Aaron Raj Bhatti

What's in this package: (1) what changed since review R1 and the status of every R1 finding, (2) whole-board LTspice simulations (review item M4), grouped by schematic sheet, (3) the updated 11-page schematic, appended after this report.

Since the R2 package (27 Oct): U10 is now the 1″ × 1″ YLPTEC URB2405YMD-20WR3 (C5369777) instead of the 2″ × 1″ URB2405LD-20WR3, same 5 V 4 A / 20 W isolated rating and 50 V 1 s surge, with a 100 µF / 16 V output bulk C38 (CX RVT1C101M0605, C81993) per its datasheet; AUX1 re-run with it (pass). The FE-URT-2 servo driver now mounts on four YIYUAN SMTSOM320BTR M3 × 2 mm SMD standoffs (H6–H9, C5301780) as on the MES3 carriers, and a simpleRTK4 Dual GPS mount has three SMTSOM365BTR M3 × 6.5 mm SMD standoffs (H10–H12, C5301796). No other circuit changes.

1. Changes since review R1
2. Status of the R1 findings
3. Open items for the reviewer
4. Simulation summary
5. Power (page 4)
6. Relays (pages 5–6)
7. Sensors (page 7)
8. Load cell (page 8)
9. Aux power (page 9)
10. IMU (page 9 → dev kit)
11. Winch (page 10)
12. Main (page 2)
13. Not simulated, and why
14. Models and assumptions
15. Schematic (11 pages)

1 · Changes since review R1

Sheet (page)	Change	Why
Relays 1-2 / 3-4 (5, 6)	U2–U5 **INA226 → INA236AIDDFR** (LCSC C5443912, TSOT-23-8). Addresses now 0x40 / 0x41 / 0x42 / 0x43 via A0 = GND / VS / SDA / SCL. A1 and VBUS are gone; the bus voltage is read on IN–. Same CONFIG 0x4327, CAL 1707, MASK/ENABLE 0x8001 and ALERT limit; bus LSB 1.6 mV, Device ID 0xA080	M1: 50 V abs max vs the INA226's 40 V (SMBJ30A clamps up to 48.4 V). The 85 V INA238 / INA228 had 0 LCSC stock
AuxPower (9)	**J15 servo 24 V: pin 1 = +24 V, pin 2 = GND_24V**, like every other power terminal	M3. The Harvester CN5 cable gets re-pinned
AuxPower (9)	**R38** (servo RX series) **100 Ω → 1 kΩ** (LCSC C11702)	L4: limits FE-URT-2 back-feed into PA3 with the module off
AuxPower (9), Main (2), Power (4)	IMU note: **3DM-CV7-AHRS on the C-Series Connectivity Board** (6212-3009). J14 9 V → board Vaux only, never Vpri; not with the USB cable at the same time. IMU load in the power budget ≈ 0.02 A at 24 V	The IMU is now known; Vpri (3.2–5.2 V) would be destroyed by 9 V
Relays 1-2 (5)	Diagnosis note: "current present" threshold **5 mA → 3 mA**; INA236 differences listed	L2: the LED alone draws 5.1–6.7 mA over 21–29 V

Power (4)	F1 / Q1 notes use the real worst case: 13 A, Q1 ≈ 2.2 W, size the drain copper for 2.2 W	L6
LoadCell (8), Main (2)	Silkscreen notes: filled box **"5V LOAD CELL – NEVER 24V"** at J12 / J13, placed away from the 24 V terminals; **"CAN – NOT ETHERNET"** at J4	M2 (silkscreen instead of a keyed connector, by choice: JLC doesn't stock the mating plugs) and L5
Footprints	**J4 RJ45 1×4 rebuilt from the HCTL drawing**: ports 14.0 mm apart (was 13.97), Ø1.5 shield holes (was Ø1.7), shield and peg positions per the drawing. U1 / U9 converter footprints checked against the JLC library and the YLPTEC drawing; U10 (URB2405YMD-20WR3) against the JLC library and the Mornsun drawing	L9, and the custom-footprint check before routing
sim/ (new)	16 LTspice benches covering every sheet with an analog or power circuit, on the real supply chain (sections 4–12)	M4

> Unchanged since R1: the GPIO map (no coil feedback; PA0/PA1 winch, PA2/PA3 servo USART2), the servo TX pull-up R37, the Winch_48V net class and isolation notes, the block diagram and the page 11 finger map. All of these were already in the R1 schematic.

2 · Status of the R1 findings

ID	Finding	Status	How
M1	INA226 40 V vs the SMBJ30A clamp	Fixed	INA236 (50 V). PWR2: the bus peaks at 40 V at D9's rated pulse
M2	24 V miswire on the load-cell terminals	Mitigated	Silkscreen box and placement away from the 24 V terminals (no keyed connector, by choice)
M3	J15 polarity	Fixed	Pin 1 = +24 V
M4	No simulation evidence	Done	16 benches, sections 4–12
L1	Relay LED resistor power	No change needed	R14 / R15 / R22 / R23 are already 1206 (0.25 W)
L2	5 mA threshold	Fixed	3 mA in the diagnosis note (firmware)
L3	Servo return through logic GND	Layout	Layout checklist: J15, J17 and NT1 together, away from the load cell
L4	PA3 back-power	Fixed	R38 = 1 kΩ; AUX3 shows 8.8 ns edges at 1 Mbit/s
L5	RJ45 looks like Ethernet	Silkscreen	"CAN – NOT ETHERNET" on the board and the panel
L6	Stale Power notes	Fixed	13 A / 2.2 W

L7	3 A PTC derating	Open	AUX1: F11 sits at its 3 A hold if the screen draws 3 A. Measure the screen; 4 A-hold PTC if needed
L8	Screen on 5 V only	Bench	Bench test at full brightness
L9	DC/DC pin numbering	Closed	U1 / U9 / U10 match the JLC library; U10 (URB2405YMD-20WR3) also the Mornsun drawing

3 · Open items for the reviewer

- **F11 (screen PTC):** keep 3 A hold, or fit 4 A now? It depends on the ARZOPA Z1C's real current (AUX1, L7).
- **Supply limits:** the pack's full-charge voltage (7S Li-ion 29.4 V or 8S LiFePO4 29.2 V) against the AB20 variant (A/B/C rated 30 V, D rated 60 V).
- **Datasheet checks: done, all pass.** URB2405S-6WR3, URB2409S-10WR3 and URB2405YMD-20WR3 all allow a 50 V input surge for 1 s (PWR2 peaks at 40 V). Q1 AGM60P90D I_{DM} −360 A (inrush 100 A). F1 0451015 melting I^2t 97.8 A^2s (inrush 0.95 A^2s).
- **Hold-up:** 4.0 ms on total power loss (PWR6), to a 9 V cut-off. The YLPTEC datasheets give an undervoltage shut-off of 5.5–6.5 V, so the real figure is a little longer. Enough for glitches; add bulk on +24V only if a clean shutdown is wanted.

4 · Simulation summary

> LTspice 17.2.4. Benches are in `MES4_Carrier_EC/sim/` and named by sheet: PWR = Power, RLY = Relays, SNS = Sensors, LC = LoadCell, AUX = AuxPower, IMU = AuxPower → dev kit, WCH = Winch, MAIN = Main. Each section below lists the sheet, the parts simulated and the bench file.

Group	Sheet (page)	Bench	Test	Result
Power	Power (4)	PWR1	XT30 hot-plug, full pack (29.4 V) via the AB20, 2 m cable	Pass
Power	Power (4)	PWR2	12.4 A 10/1000 µs pulse (D9 rating)	Pass
Power	Power (4)	PWR3	IEC 61000-4-5 0.5 kV surge	Not applicable
Power	Power (4)	PWR4	Battery connected backwards	Pass
Power	Power (4), fed by the AB20	PWR6	Line changeover and hold-up	Pass
Power	Power (4) + Main (2)	PWR5	+5V_24 / module +5V under load steps	Pass
Relays	Relays 1-2 / 3-4 (5, 6)	RLY1	SRD coil driver on / off	Pass
Relays	Relays 3-4 (6)	RLY2	Solenoid output switched off	Pass
Sensors	Sensors (7)	SNS1	PC817 input, 20 V sensor, CTR 50 %	Pass
Load cell	LoadCell (8)	LC1	ADC input filter, AC response	Pass
Aux power	AuxPower (9)	AUX1	U10 start-up into the screen	Pass, check F11
Aux power	AuxPower (9)	AUX2	3 A servo stall step	Pass
Aux power	AuxPower (9)	AUX3	Servo UART, 1 Mbit/s	Pass
Winch	Winch (10)	WCH1	IM03GR coil driver on / off	Pass
IMU	AuxPower (9) → J14 → dev kit	IMU1	SW2 switches 9 V into the CV7 dev board	Pass
Main	Main (2) + Relays (5, 6)	MAIN1	I²C1 at 400 kHz, ALERT	Pass

> Everything still to decide or check is collected in section 3.

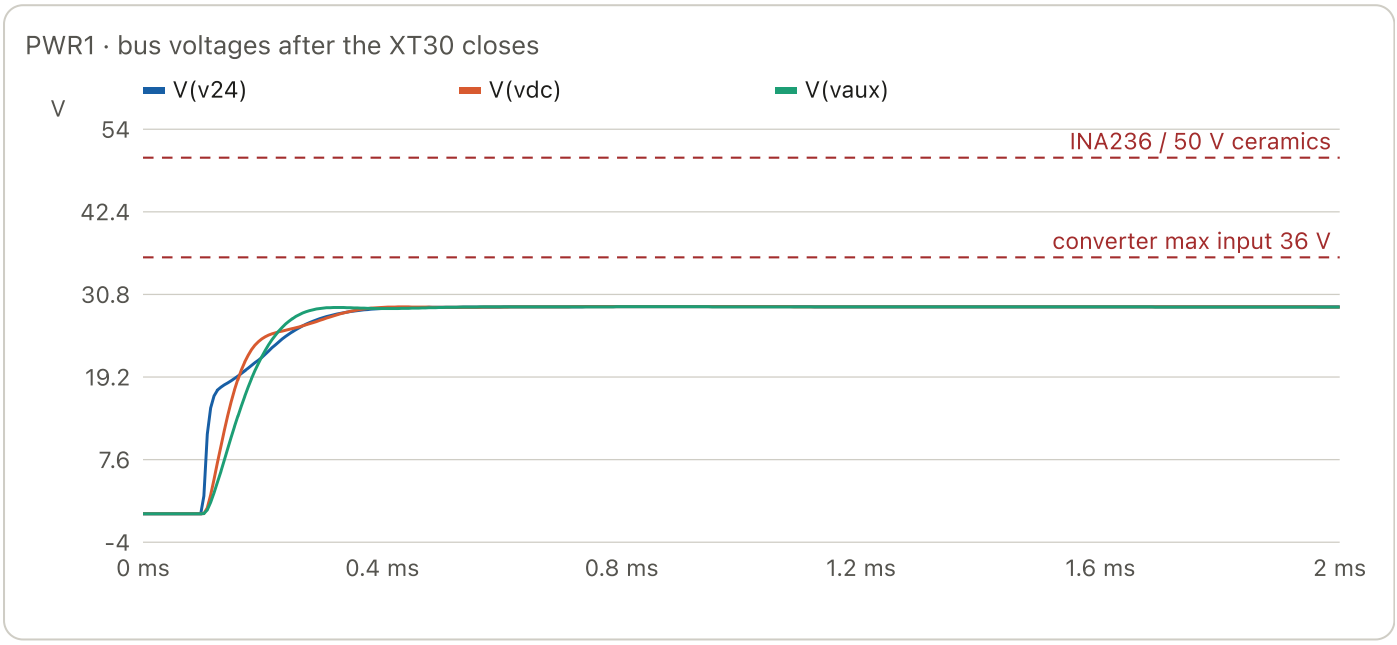
5 · Power · sheet Power (page 4)

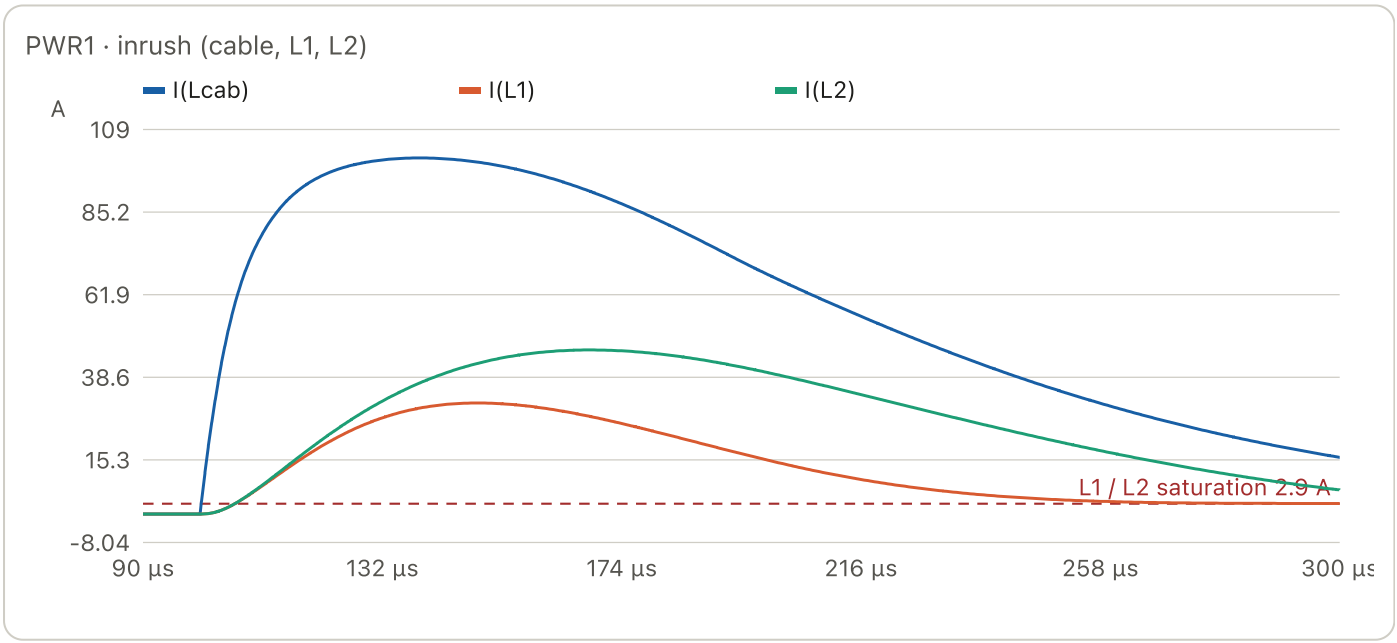
Supply into J5 (your system): 48 V main battery → Mean Well **DDR-480C-24** (set to 24 V; adjustable 24–28 V; OVP 28.8–35 V; 4 kV isolated; 20 A) → **AB20** line 1. 24 V 12 Ah Li-ion pack → **XY-CD63L** discharge monitor → AB20 line 2. The AB20 is a diode-OR ("zero-delay", anti-backflow): the higher input feeds the carrier. Model: `source_ab20.inc`.

Parts: J5 XT30, F1 15 A, Q1 AGM60P90D + D8/R6, D9 SMBJ30A, C1–C3 · L1 + C4/C5 → U1 URB2405S-6WR3 (+24V_DC) · L2 + C26–C30 → U9 / U10 (AuxPower) · F9 + C31/C32 (servo). Shared model: `common.inc`.

PWR1 · hot-plug (`pwr1_hotplug.net`)

The XT30 closes on the AB20 output with the pack fully charged (29.4 V, above the DDR's 24 V, so the pack feeds the carrier), through a 2 m cable (2 µH, 33 mΩ). +24V rises cleanly to 29.1 V, with no overshoot on any branch, so D9 never conducts. Charging about 450 µF pulls a current pulse peaking at 100 A and lasting about 0.1 ms (F1 I^2t 0.95 A^2s), so the XT30 will spark. L1 and L2 saturate briefly (31 / 46 A in this linear model). That's harmless: Q1 AGM60P90D is rated −360 A pulsed (AGMSEMI datasheet), and F1 0451015's nominal melting I^2t is 97.8 A^2s (Littelfuse), about 100× the simulated 0.95 A^2s.

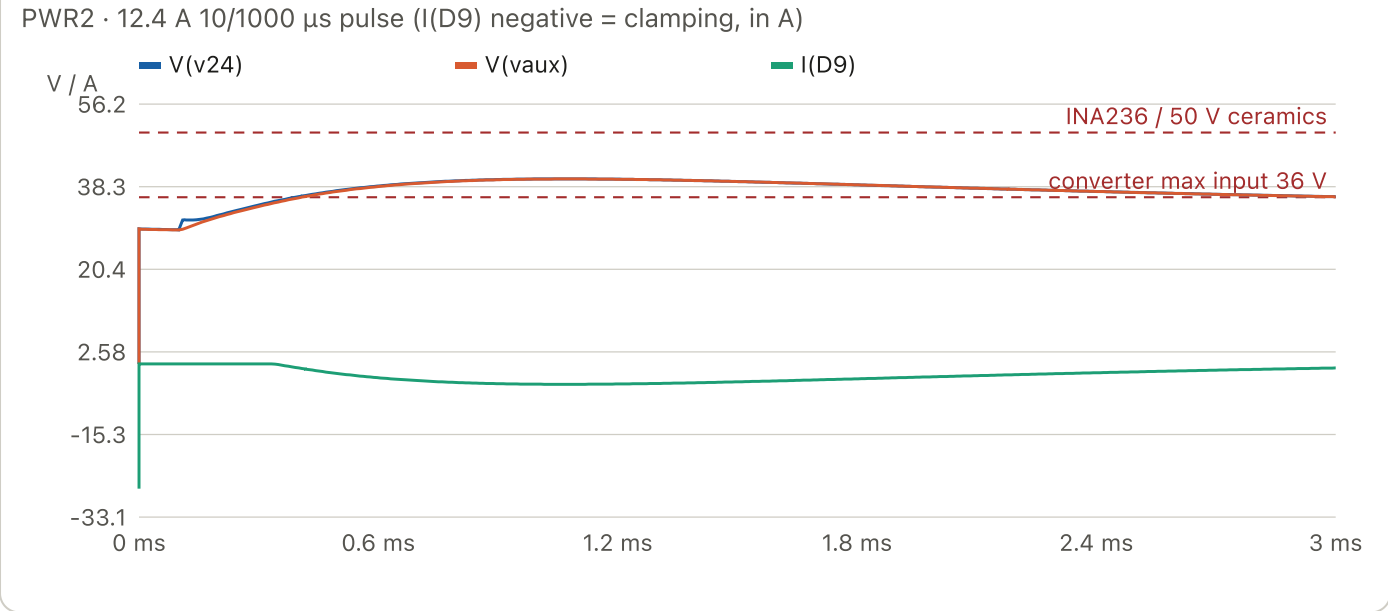




PWR2 · surge at D9's rating (`pwr2_tvs_pulse.net`)

This is the design-limit case: both AB20 inputs are gone (bus held at 29.2 V by a weak source) and a 12.4 A, 10/1000 µs pulse arrives on the cable. With the AB20's diodes blocking any return path, this is also what servo back-EMF would face. The bulk capacitors take most of it: the bus peaks at 40.0 V and D9 takes 4.4 A and 0.31

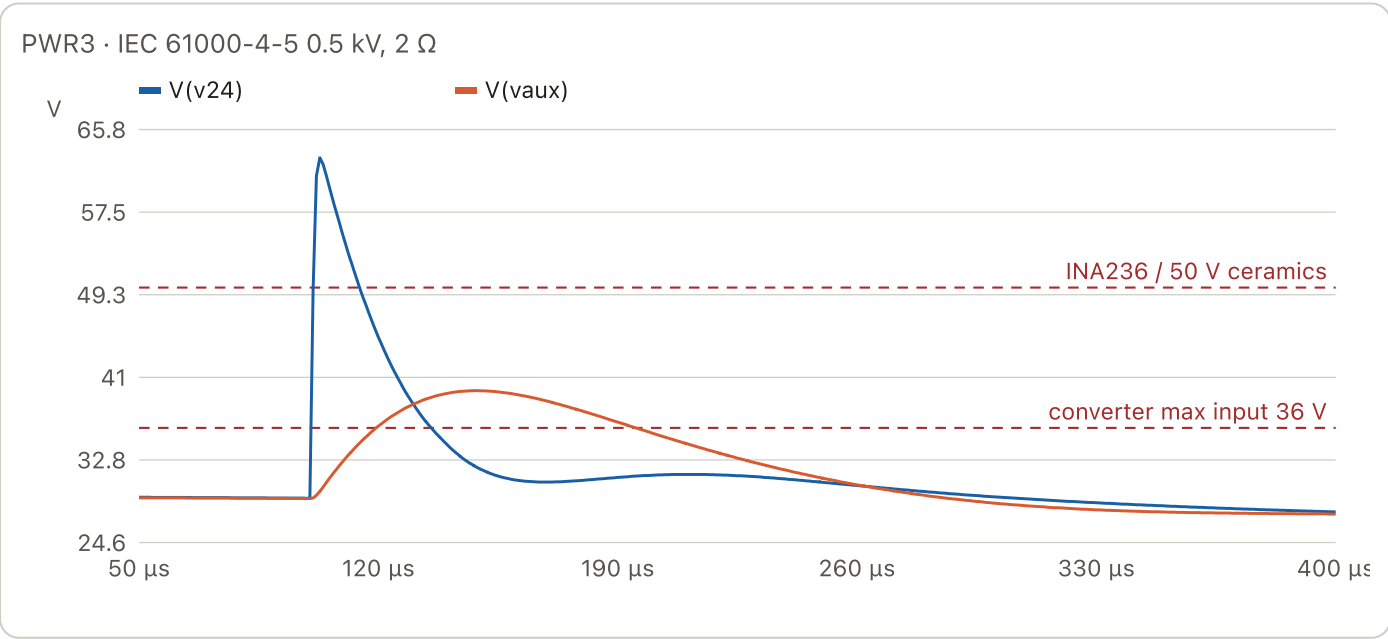
J of its ≈ 0.87 J rating. That leaves 10 V of margin to the INA236. U1, U9 and U10 see 40 V for about 1 ms: above their 36 V operating range, but inside the 50 V for 1 s input surge rating on all three YLPTEC datasheets.



PWR3 · IEC 61000-4-5 surge (`pwr3_iec_surge.net`) · not applicable

Not applicable to this system: the carrier's 24 V comes from the isolated DDR-480C-24 and an internal pack through the AB20, on short wiring inside the robot. The run is kept for reference, in case the bus ever gets an external charger port or a long harness.

0.5 kV line-to-line, 1.2/50 µs, 2 Ω, through an 18 µF coupling capacitor, with the battery behind a 1.5 mH decoupling inductor. D9 conducts 27 A and the bus reaches 62.9 V, above the INA236's 50 V and the 50 V ceramics. +24V_AUX stays at 39.7 V behind L2. This surge is normally required only for long or external DC power cables. If the bus ever goes off the robot (a charger port, a long harness), fit an **SMCJ30A** (1500 W, same standoff, Vc 48.4 V at 31 A, about 47 V here).

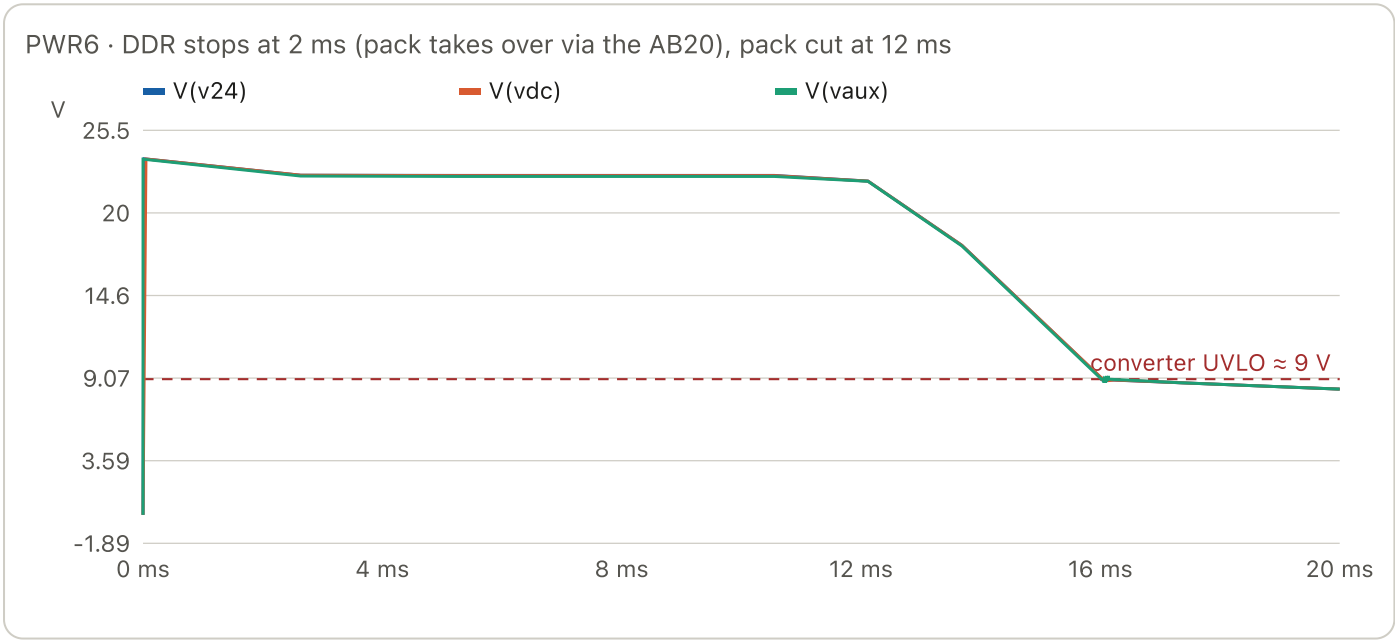


PWR4 · reverse battery (`pwr4_reverse.net`)

The battery is connected as −29.2 V. Q1's body diode is reverse-biased and its gate stays off, so +24V stays between −4 mV and +2 mV and only 0.12 mA of leakage flows. D9 sits after Q1, so it never sees the reverse voltage. Pass

PWR6 · line changeover and hold-up (`pwr6_changeover.net`)

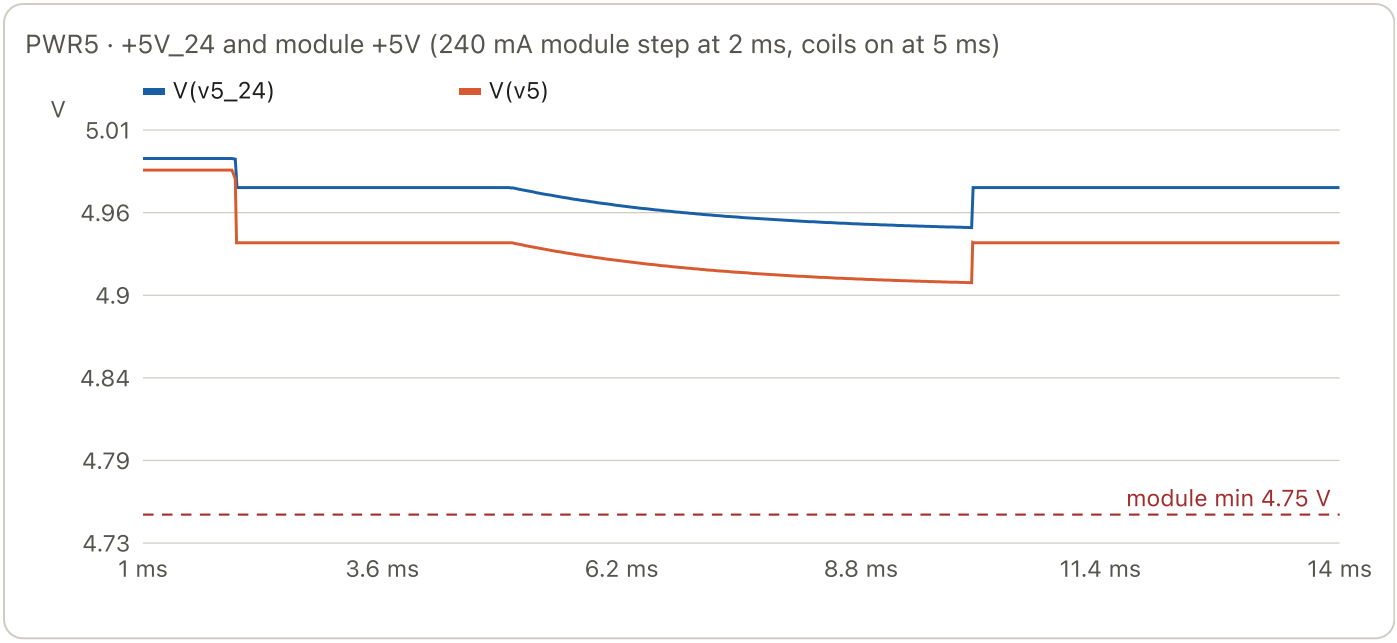
The DDR on line 1 carries the load at 24 V (the carrier sees 23.6 V after the AB20 diode), with the pack part-discharged at 23.0 V. At 2 ms the DDR stops (48 V battery below its 33 V UVLO, or an OVP trip). The pack takes over through the AB20 with no gap: +24V simply steps to 22.5 V. At 12 ms the XY-CD63L cuts the pack as well. With both sources gone, the carrier keeps running on its own bulk capacitance for **4.0 ms** (screen at full load) before U1's input falls to 9 V and the module loses 5 V. That's enough to ride through glitches, not enough to shut down cleanly. If you ever want a clean shutdown on total power loss, add bulk capacitance on +24V (2,200 µF would give about 20 ms). Also, the DDR alone can never put more than 35 V on the bus (its OVP), which is inside the converters' 36 V and the INA236's 50 V.



PWR5 · 5 V rails (`pwr5_5v_rail.net`, sheets Power + Main)

Parts: U1 output → C6/C7 (+5V_24) → F2 1.1 A PTC → FB1 → C8/C9 → module fingers 20/22 (module 10 µF + SMF6.0A); relay K1–K4 and winch K5 coils on +5V_24; load cell excitation via FB2.

The module steps from 40 mA to 240 mA at 2 ms, then four relay coils and one winch coil switch on together at 5 ms (0.27 A after 5 ms, rising to 0.31 A). The module's +5V stays between 4.91 V and 4.99 V, inside its 4.75–5.5 V window, and +5V_24 never drops below 4.95 V.



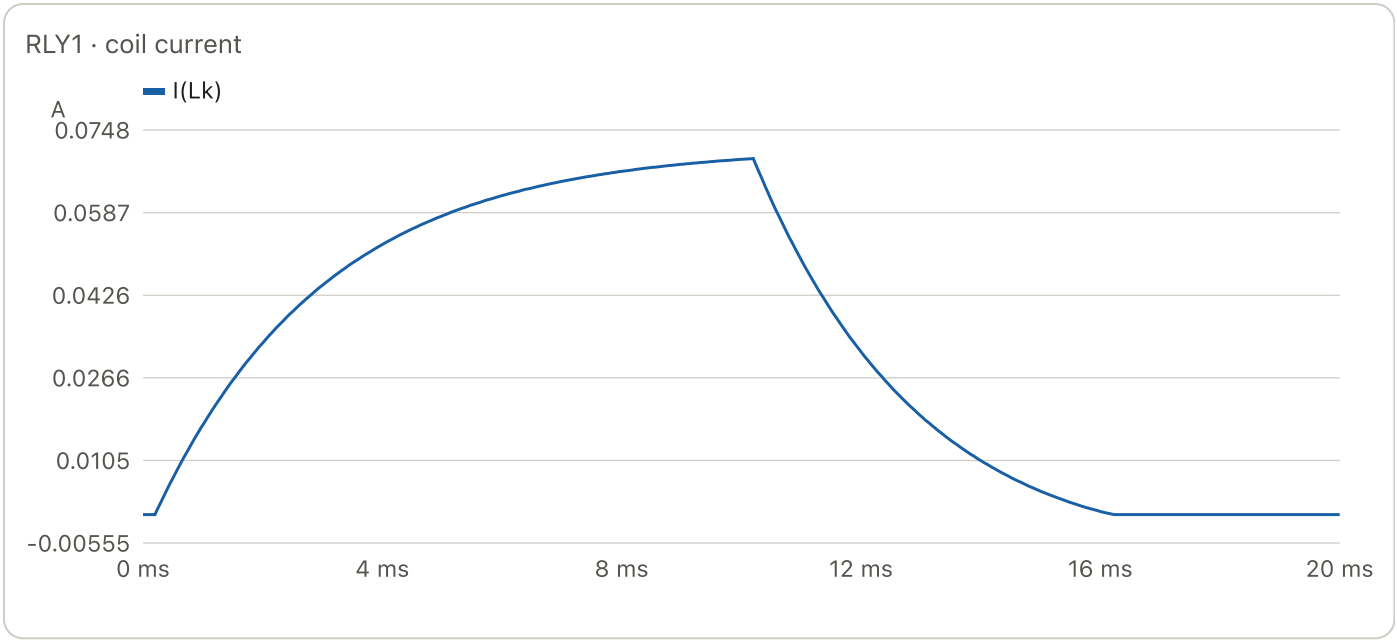
6 · Relays · sheets Relays 1-2 / Relays 3-4 (pages 5–6)

Parts per channel: GPIO → R9 100 Ω (R10 100k pull-down) → Q2 AO3400A → K1 SRD-5VDC-SL-A coil, D10 1N4148W flyback; contact → F3 3 A PTC → J6, D12 SS34 load flyback, LED4 + R14; R8 30 mΩ shunt → U2 INA236. Same for channels 2–4.

RLY1 · coil driver (`rly1_coil_driver.net`)

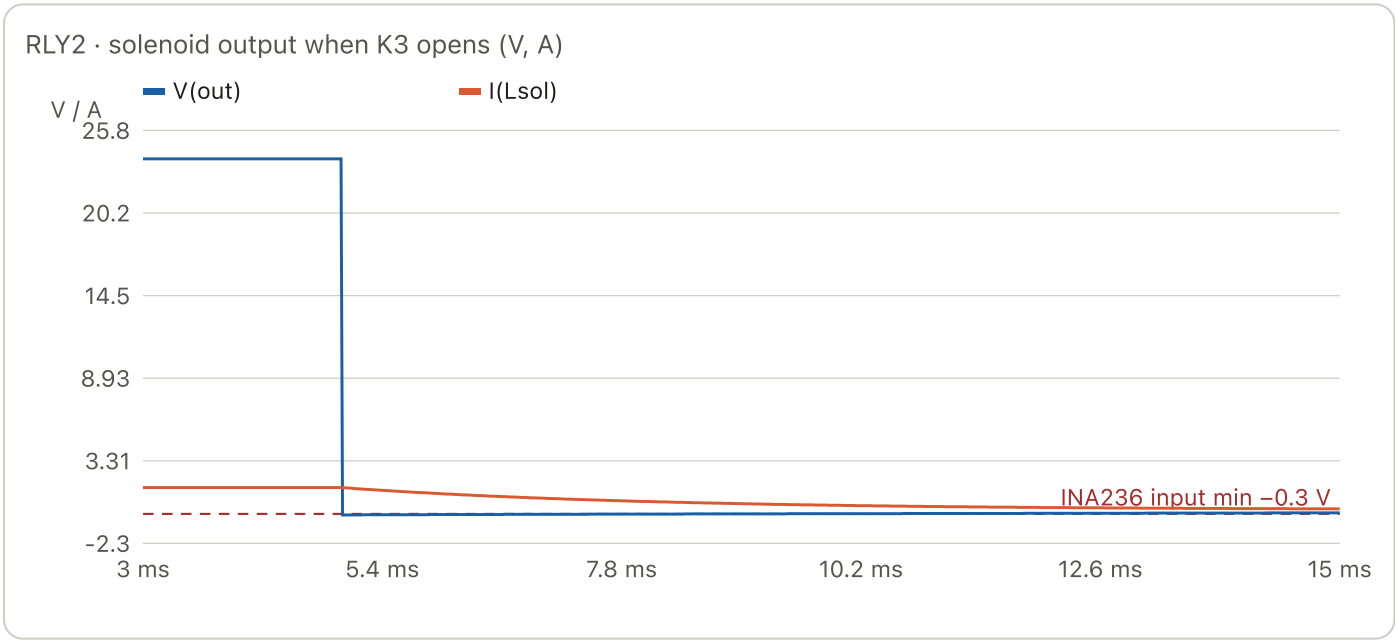
3.3 V GPIO on at 0.2 ms, off at 10.2 ms. The gate rises in 0.28 µs, and the coil draws 69 mA. At turn-off, D10 clamps the drain at 5.8 V (the AO3400A is rated 30 V). The coil current takes 3.9 ms to fall to 10 mA, which delays the relay's release a little. That's normal with a plain diode flyback and fine for lamps and the solenoid.





RLY2 · solenoid switched off (`rly2_solenoid_off.net`, channel 3 battery stopper)

A 24 V, 1.5 A solenoid (16 Ω, 50 mH assumed) is on when K3 opens at 5 ms. D16 SS34 clamps the output at −0.36 V and carries 1.49 A (rated 3 A). The current decays to 10 % in 6.9 ms. The INA236's IN− sits on the supply side of the contact and stays at 23.96 V, so the flyback never reaches its −0.3 V input limit.

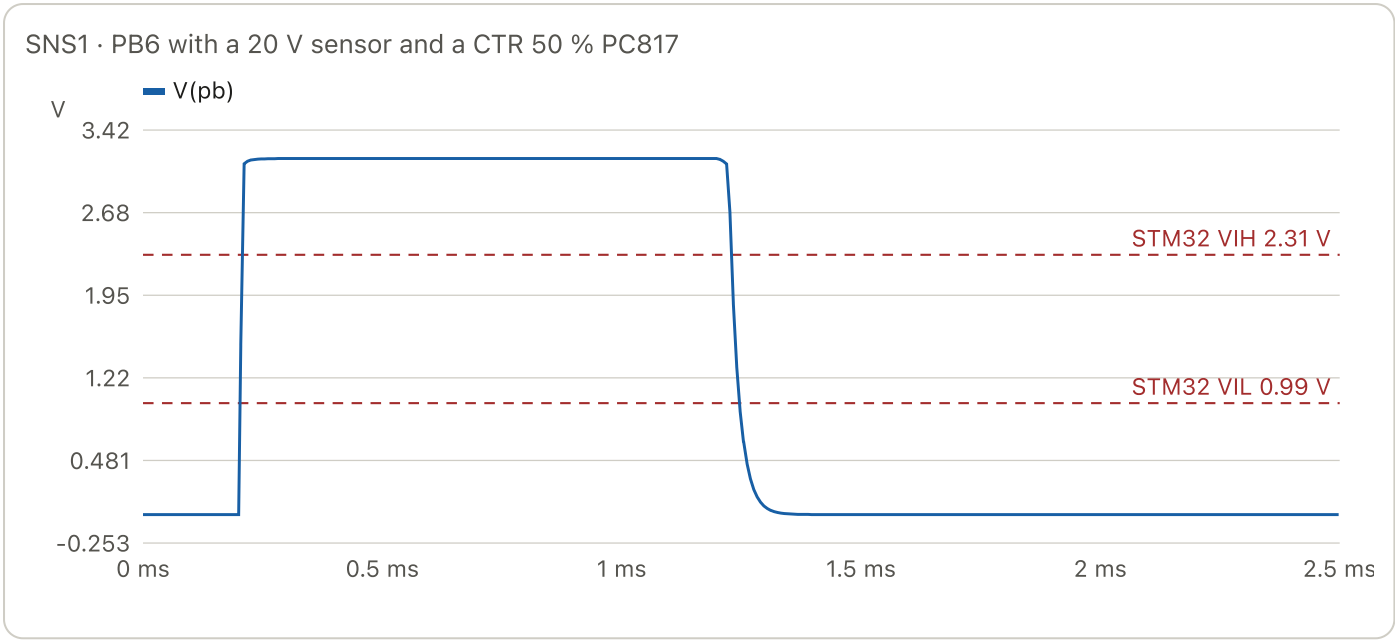


7 · Sensors · sheet Sensors (page 7)

Parts: J10 → F7 100 mA PTC → sensor; sensor output → R24 10k → U6 PC817 (D18 reverse clamp) → PB6 with R25 10k pull-down, collector on +3V3_MOD. Sensor 2 (PC15) is identical.

SNS1 · PC817 input (`sns1_pc817.net`)

This is the weak case: the sensor output is only 20 V (low battery) and the PC817 has its minimum CTR of 50 %. The LED draws 1.84 mA. PB6 goes HIGH to 3.17 V, well above the STM32's 2.31 V threshold, with about 3× margin on CTR. Rise is 4 µs and fall is 34 µs, much faster than any proximity sensor switches.

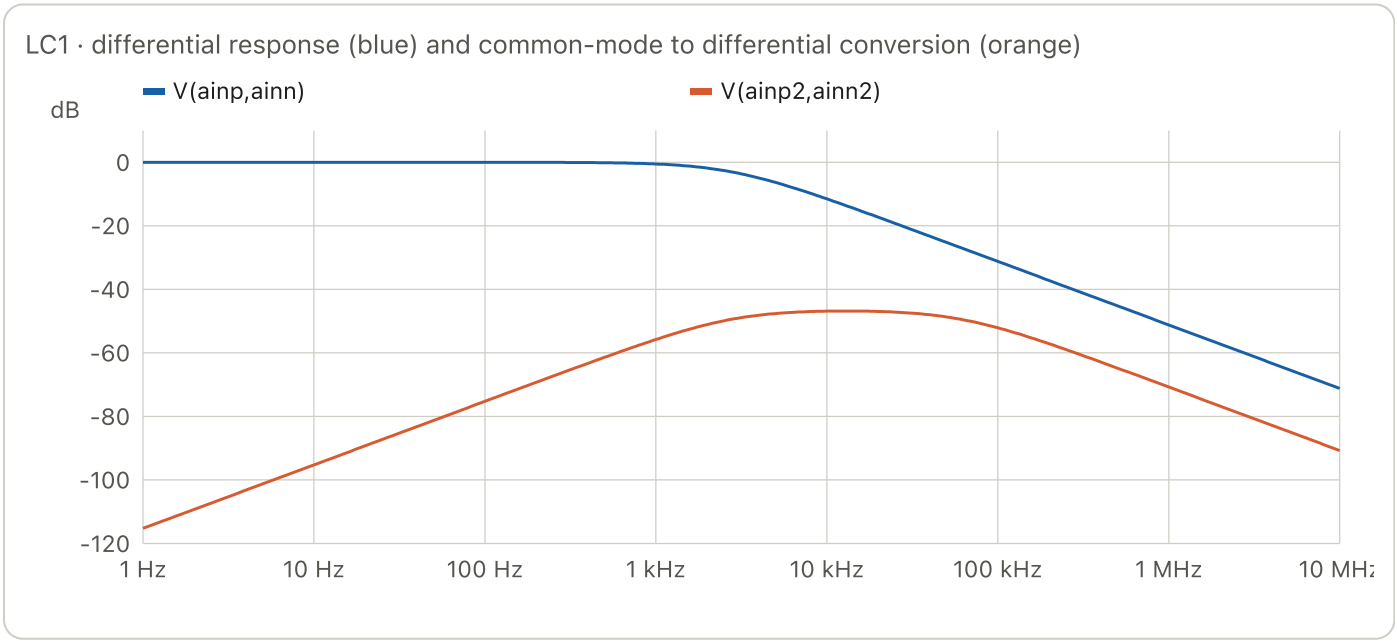


8 · Load cell · sheet LoadCell (page 8)

Parts: J12 SIG → R28/R29 100 Ω → C17 100 nF differential + C16/C18 10 nF to GND → D20/D21 BAV199 clamps → U8 ADS1232 (gain 128, 10 SPS). Source: two SOPLF-102 cells in parallel (700 Ω each → 350 Ω).

LC1 · input filter (`lc1_input_filter.net`, AC)

The differential signal is flat in the passband (−0.001 dB at 50 Hz) and rolls off at −3 dB at 2.76 kHz, including the cells' 350 Ω source. Common-mode noise converts to a differential error only through component mismatch. With C18 10 % low, that's −56 dB at 1 kHz, worst ≈ −47 dB around 20 kHz and −71 dB at 1 MHz. The ADS1232's 10 SPS digital filter removes what's left at those frequencies. The C0G capacitors keep the mismatch small and stable.

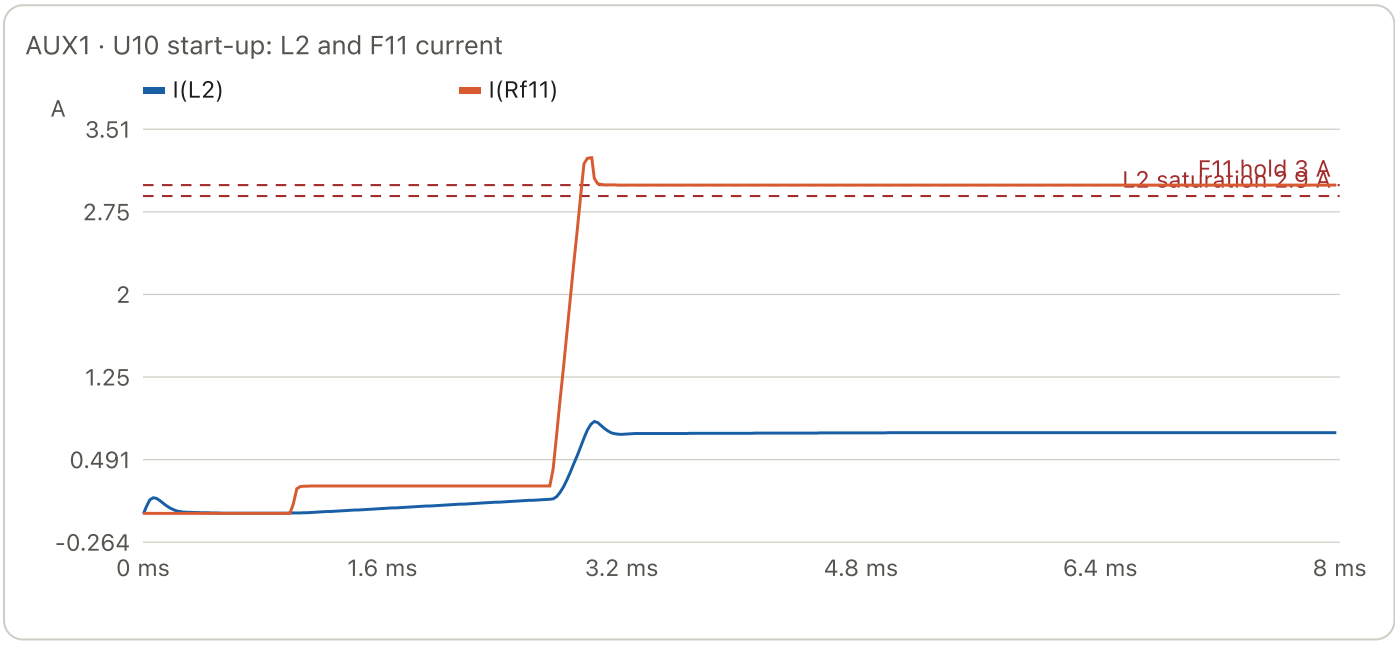


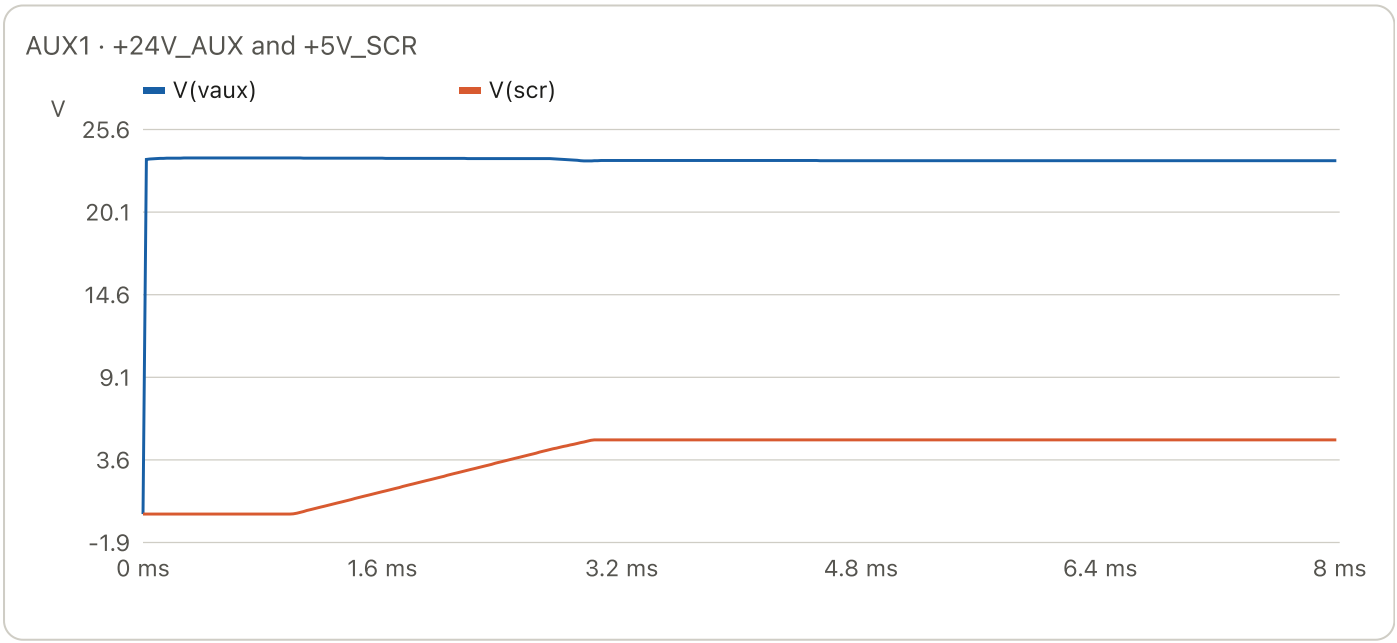
9 · Aux power · sheet AuxPower (page 9)

Parts: L2 + C26/C27/C28/C30 → U9 URB2409S-10WR3 (IMU 9 V) and U10 URB2405YMD-20WR3 (screen 5 V) → C33/C34/C38 → F11 3 A PTC → J16 USB-C · F9 3 A PTC + C31/C32 → J15 servo 24 V · servo UART J17: R39 100 Ω / R38 1k series, D23/D24 ESD, R36/R37 10k pull-ups.

AUX1 · U10 start-up (`aux1_u10_startup.net`)

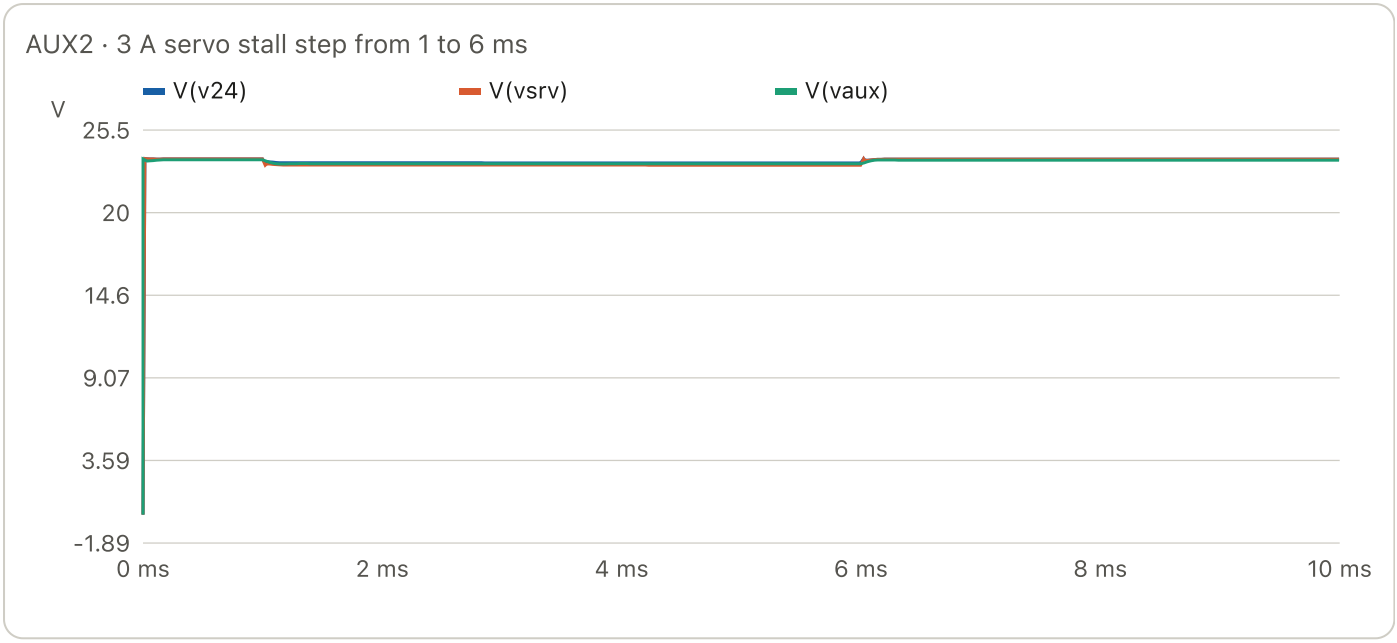
U10 is now the 1″ × 1″ URB2405YMD-20WR3 (9–36 V in, 40 V max, 50 V for 1 s; 5 V 4 A; 88 % minimum efficiency; over-current trip 110–190 % of 4 A, hiccup). With the bus up, it soft-starts its 5 V output in 2 ms (fast-case assumption; the datasheet's typical start-up is 10 ms, which is gentler) into C33/C34, the new 100 µF C38 (the datasheet's Cout), F11 and the screen, which pulls the full 3 A. U10 peaks at 3.58 A while charging the output capacitors: 0.8 A under the lowest possible over-current trip (4.4 A), so it can't go into hiccup at start-up. It then runs at 3.0 A (15 W, 75 % of its 20 W rating) with +5V_SCR at 4.94 V, inside its ±3 % tolerance. The total output capacitance (≈ 240 µF with the screen) is far below its 10,000 µF limit, and at 75 % load it needs no heatsink up to about 70 °C. L2 carries 0.84 A in total (U9 with the IMU + U10), far below its saturation current, and +24V_AUX dips only to 23.5 V. F11 peaks at 3.25 A and sits at 3.0 A, exactly its 23 °C hold current. In a warm enclosure (≈ 2.2 A hold at 60 °C) it would trip if the screen really draws 3 A.





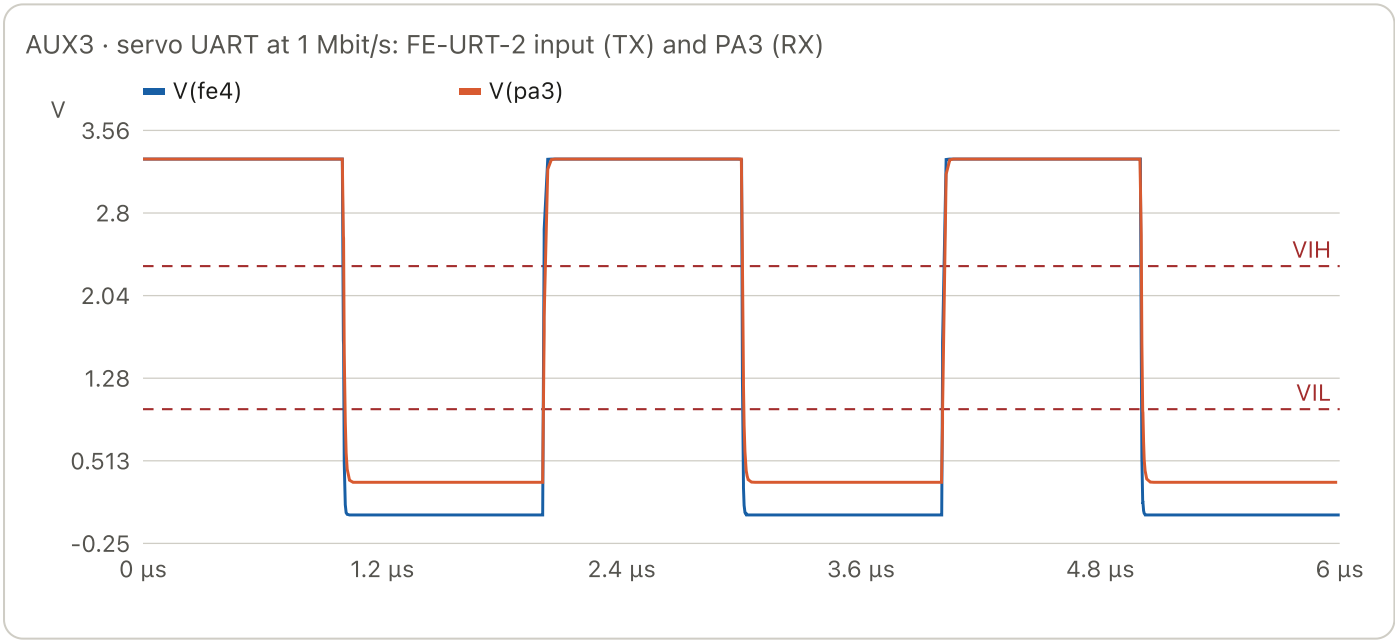
AUX2 · servo stall step (`aux2_servo_step.net`)

The servo draws 3 A for 5 ms (stall), supplied through the AB20 from the DDR at 24 V over a 2 m cable. +24V_SRV dips to 23.19 V and the main +24V to 23.32 V. The relays, INA236s and +24V_AUX (23.27 V) barely notice.



AUX3 · servo UART (`aux3_servo_uart.net`)

1 Mbit/s (Feetech default) over 0.3 m, with the FE-URT-2 at 3.3 V levels. TX edges at the FE-URT-2 rise in 4 ns. RX edges at PA3 rise in 8.8 ns through the 1 kΩ R38: both under 1 % of the 1 µs bit. PA3's LOW level is 0.31 V (1k against the 10k pull-up), under the STM32's 0.99 V threshold.



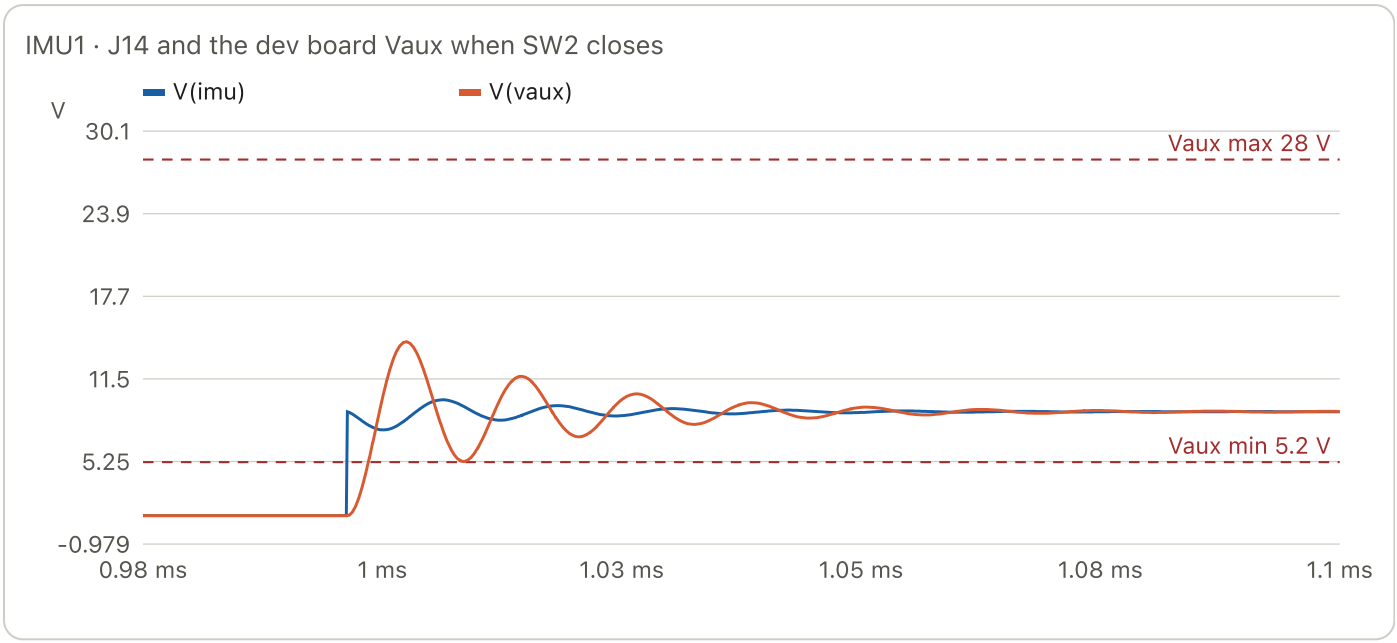
10 · IMU · sheet AuxPower (page 9) → J14 → MicroStrain dev kit

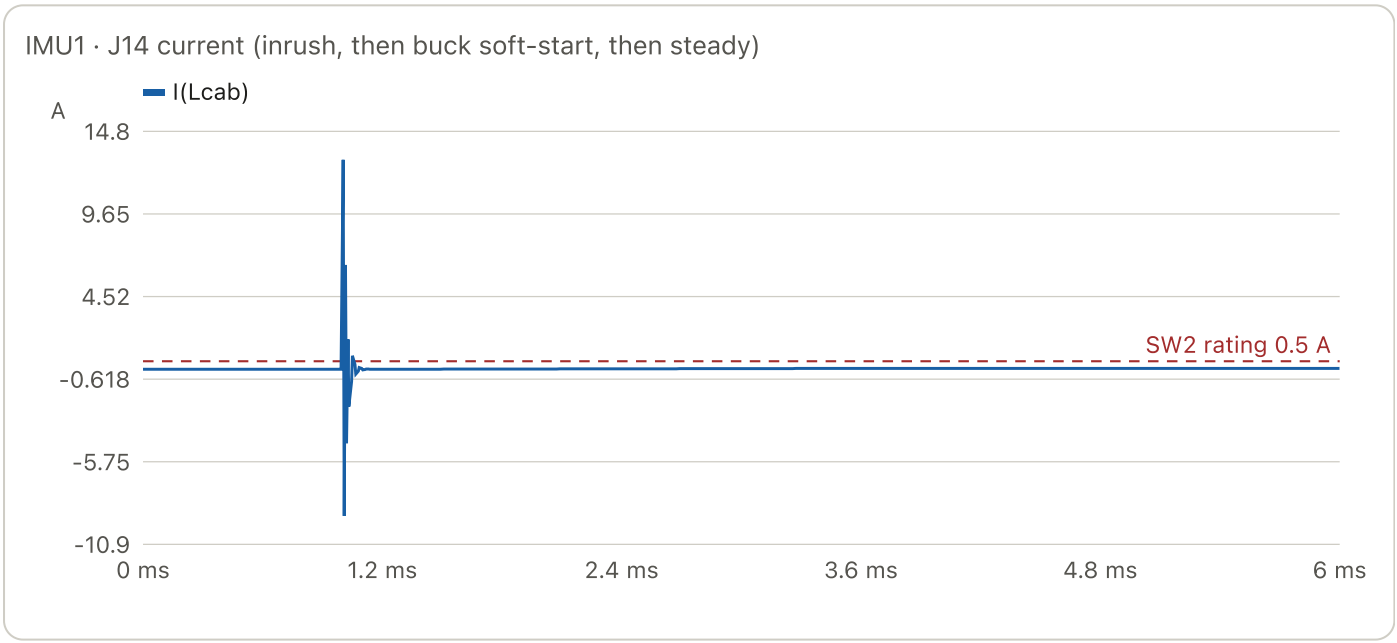
Parts: U9 URB2409S-10WR3 9 V → C29 → SW2 SS12D10G5 → LED8 / R35 → J14 → 1 m cable → **C-Series Connectivity Board** (part of the 6212-3009 USB development kit) Vaux input → its TPS54260 buck → 5 V → **3DM-CV7-AHRS** (3.2–5.2 V, 0.23 W, per HBK's product page and the board's Interface Control Document).

IMU1 · switching the IMU on (`imu1_devkit_switch_on.net`)

SW2 closes with U9's 9 V already up. The 1 m cable and the board's ceramic input capacitors (2 × 2.2 µF + 0.1 µF) ring: Vaux overshoots to 14.3 V, inside the board's 5.2–28 V Vaux range. The inrush through SW2 peaks at 9.1 A for a few microseconds (0.3 mJ of capacitor charge). Then the buck soft-starts, and the steady draw is 44

mA at J14, 48 mA through SW2 with LED8. That's about a tenth of SW2's 0.5 A rating, so review item L8's SW2 check is closed. U9 (1.11 A) runs at about 4 % load.





Wiring rules from the dev kit's Interface Control Document, not things a simulation can show:

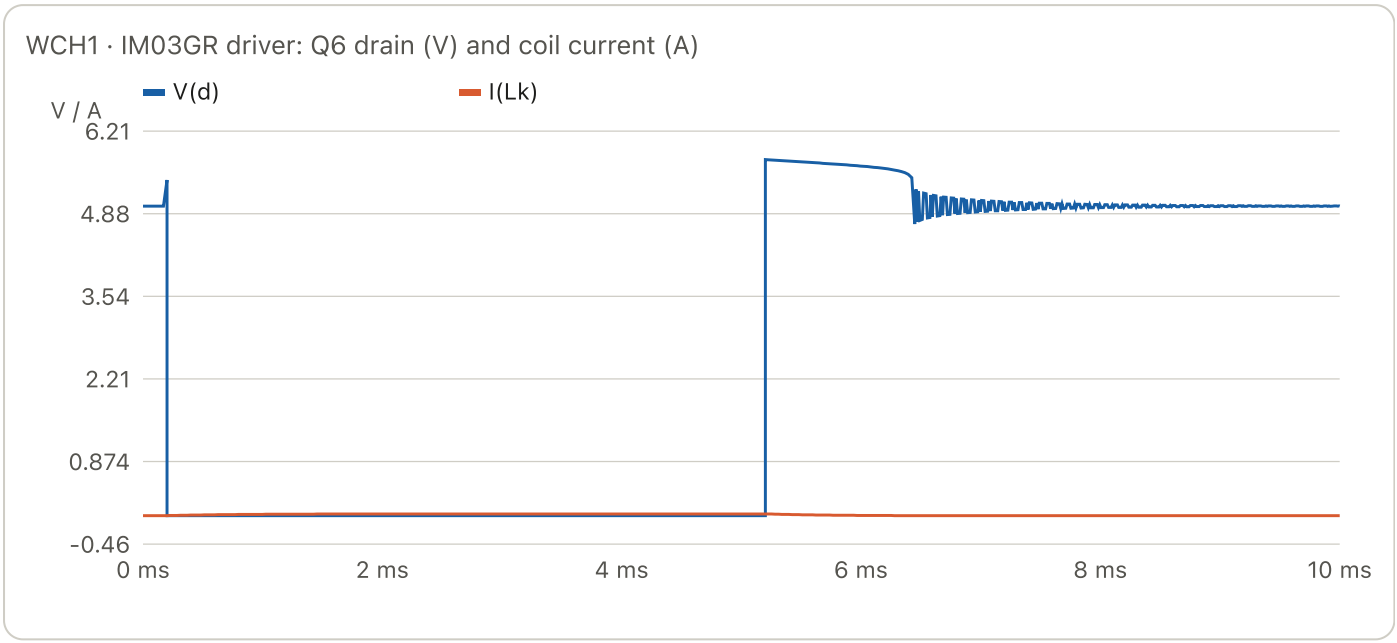
- **Feed J14's 9 V into the board's Vaux input** (micro-DB9 pin 6, "+4 V to +35 VDC" in the pin table, 5.2–28 V in the schematic notes). **Never into Vpri** (pin 3, 3.2–5.2 V): Vpri goes straight to the CV7, and 9 V there would destroy it.
- **Vaux or Vpri, never both.** The USB dev cable (micro-DB9 to USB) powers Vpri from the PC's 5 V. When the USB cable is plugged in for configuration, switch SW2 off.
- J14 pin 2 is the carrier's logic GND. Over USB the IMU also connects to the PC's ground: normal, but keep the IMU's USB and power cables together.

11 · Winch · sheet Winch (page 10)

Parts: PA0 → R43 100 Ω (R44 100k) → Q6 AO3400A → K5 IM03GR coil (178 Ω), D25 1N4148W flyback. K6 / PA1 is identical. The contact side (J18) is isolated and passive.

WCH1 · coil driver (`wch1_coil_driver.net`)

The coil draws 28 mA. At turn-off, D25 clamps the drain at 5.75 V, and the current falls to 3 mA in 0.87 ms, so release is quick.

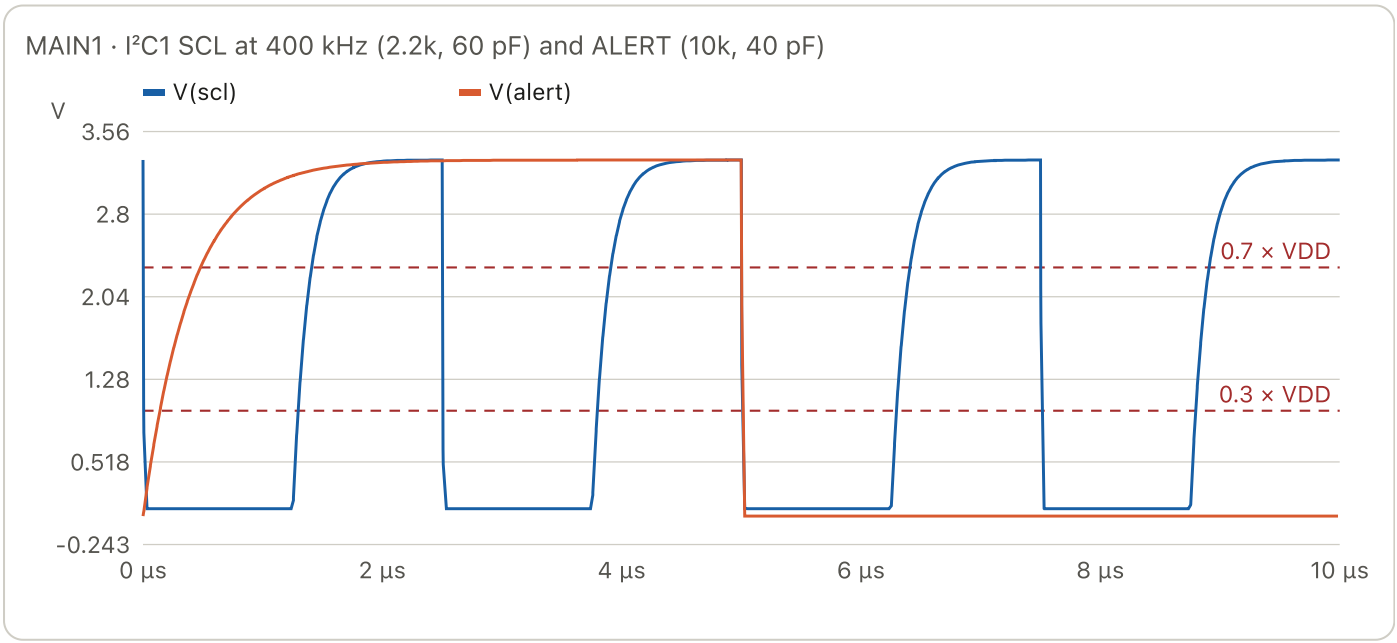


12 · Main · sheet Main (page 2), devices on Relays (pages 5–6)

Parts: R3/R4 2.2k I²C1 pull-ups and R5 10k ALERT pull-up to +3V3_MOD; four INA236 (U2–U5) and the STM32 on the bus.

MAIN1 · I²C and ALERT (`main1_i2c.net`)

400 kHz with an estimated 60 pF of bus capacitance. SCL rises (30–70 %) in 112 ns, against the 300 ns fast-mode limit, and the LOW level is 0.09 V. ALERT (10k, 40 pF) rises in 0.34 µs, which is plenty for an interrupt or polled line.



13 · Not simulated, and why

Sheet (page)	Circuit	Reason
CAN (3)	DB9 + RJ45, H24VNT2B ESD	Passive on the carrier: the transceiver and terminator are on the MES4-MCU module, which has its own review
AuxPower (9)	USB-C CC (R40/R41), shield R42/C37	Static: 10k Rp to 5 V advertises 3 A; nothing changes over time
LoadCell (8)	ADS1232 digital interface (PA4–PA6)	Digital at 10 SPS, handled in firmware
Winch (10)	Contact interlock, J18	Logic of the contacts, traced in the design review; isolated 48 V side
Main (2)	ST-Link, reset, ESD	Passive

14 · Models and assumptions

- Values come from the schematic netlist. Ceramic 10 µF / 50 V parts are modelled at ≈ 4 µF at 24–30 V, and 22 µF / 25 V at ≈ 15–18 µF at 5 V (DC bias).
- Electrolytic ESR: 100 µF / 63 V 0.25 Ω, 47 µF / 100 V 0.6 Ω (datasheet-class, not measured).
- Supply: DDR-480C-24 at 24 V (10 mΩ); pack 29.4 V full (7S Li-ion; 8S LiFePO4 would be 29.2 V), 60 mΩ assumed; XY-CD63L switch 5 mΩ assumed; AB20 Schottky diodes ≈ 0.45 V at 10 A (assumed; Autoro gives no drop figure). Cable from the AB20 to the carrier 2 m (2 µH, 33 mΩ).
- SMBJ30A: Zener model fitted to Vc 48.4 V at 12.4 A. AO3400A: generic VDMOS (Vth 1.05 V). 1N4148W and SS34: generic diode models. PC817: behavioural, CTR 50 %, 20 µs pole.
- Converters are constant-power loads above 9 V (U1 3.75 W, U9 0.5 W for the CV7 dev kit, U10 17 W), with 5 ms soft start (U10's output ramp 2 ms in AUX1). Screen input 100 µF, drawing 3 A.
- Assumed inductances (not on the datasheets): SRD coil 0.2 H, IM03GR 0.1 H, battery-stopper solenoid 50 mH. Inductors are linear (no saturation).
- Bus capacitance: I²C 60 pF, ALERT 40 pF, PA3 7 pF. UART cable 0.3 m (100 Ω, 1.5 ns).

15 · Schematic

The updated schematic (11 A3 pages, KiCad 9.0.2, ERC 0 errors / 0 warnings, 198 parts) follows on the next pages.

Rerun: `python3 run_sims.py && python3 make_report.py` in `sim/`. Open any `.net` file in LTspice to see every waveform; `results.txt` has the raw measurements.